

Winter Storm Fern Situation Report

January 2026

Formatted based on LA Fire [Situation Report](#)

Summary

A major American winter storm, often referred to as Winter Storm Fern, started on Friday, January 23 and continued through January 26th. The storm brought heavy snow and freezing rain to several U.S. states, ranging from the southern plains to the Eastern United States. Several states experienced network outages and extremely cold temperatures.

The following visualizations are developed based on Meta AI for Good's data through Facebook. The data covers network coverage and population density from January 27 to February 10, with a focus on Tennessee and Kentucky and daily updates during the collection period.

Definitions

- **Baseline Ratio:** Ratio of people using the Facebook app who have turned on the Location Services device setting on their mobile device that are present in a specific tile during the baseline period over users in all tiles in the boundary box defining the crisis
- **Crisis Ratio:** Ratio of people using the Facebook app who have turned on the Location Services device setting on their mobile device that are present in a specific tile during the 8-hour window over users in all tiles in the boundary box defining the crisis
- **Population Change Density (%):** Percent change in Facebook Location Services-enabled user population counts per tile, calculated by dividing the difference by the baseline (plus a small value, usually 1)
- **Bing Tiles:** equivalent to roughly 2.4 kilometers on a side near the equator or the size of 6 x 6 city blocks

- **Administrative Regions:** the polygon boundaries for administrative regions derived from the Database of Global Administrative Areas, also known as GADM.
- **Time Windows:** aggregated over 3 fixed 8-hour time windows, beginning at 00:00, 08:00 and 16:00, that represent changes in population density

Bing Tiles vs. Administrative Regions

The Facebook Population During Crisis (Administrative Regions) dataset is equivalent to the Facebook Population During Crisis (Tile Level) dataset except instead of aggregating user counts over grid tiles, it aggregates user counts within administrative boundaries such as states, provinces and counties.

Movement

% Population Density Change

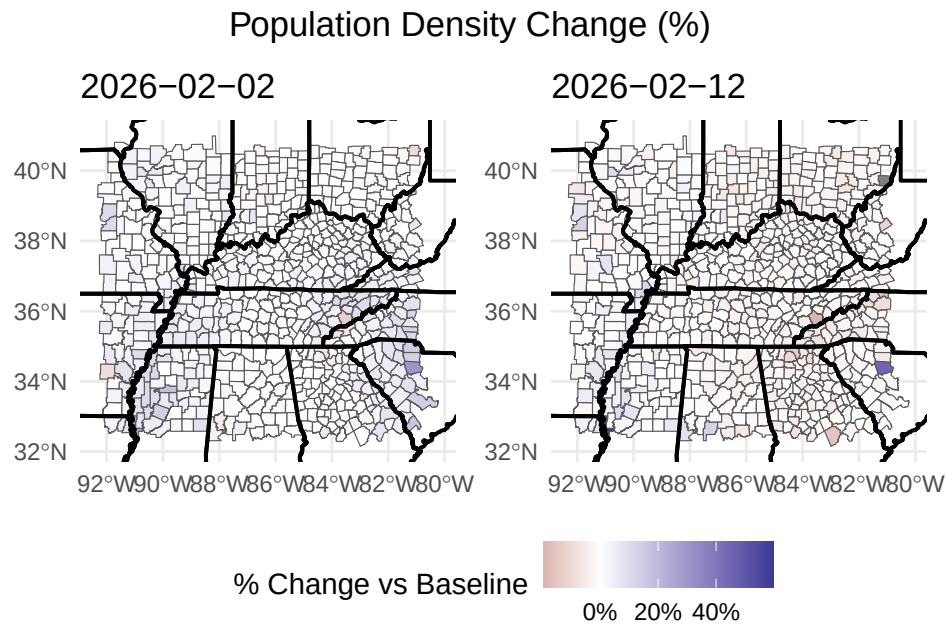


Figure 1: Based on the Bing tiles, there are only a few counties with strong population density changes.

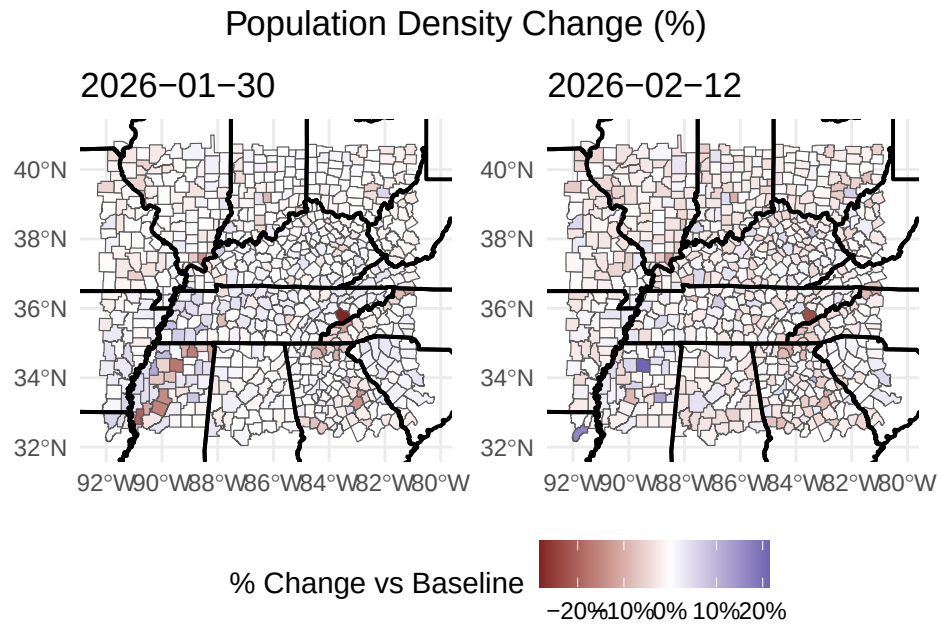


Figure 2: Based on the administrative regions, there are two counties with strong population density changes, one positively and one negatively.

Time Series of Movement

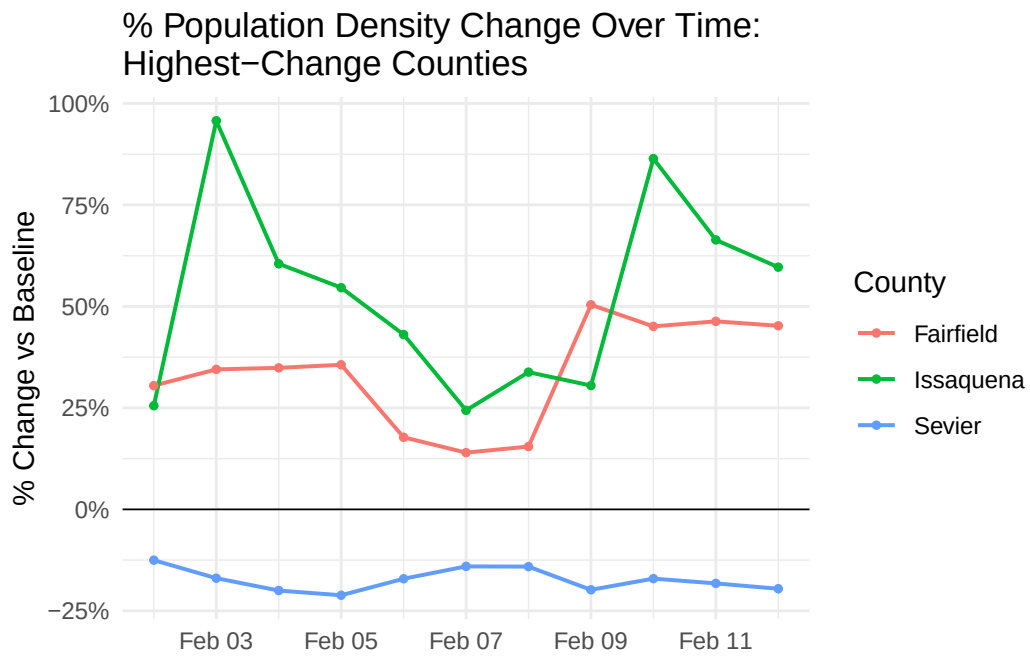


Figure 3: There are strong difference in population changes across different counties

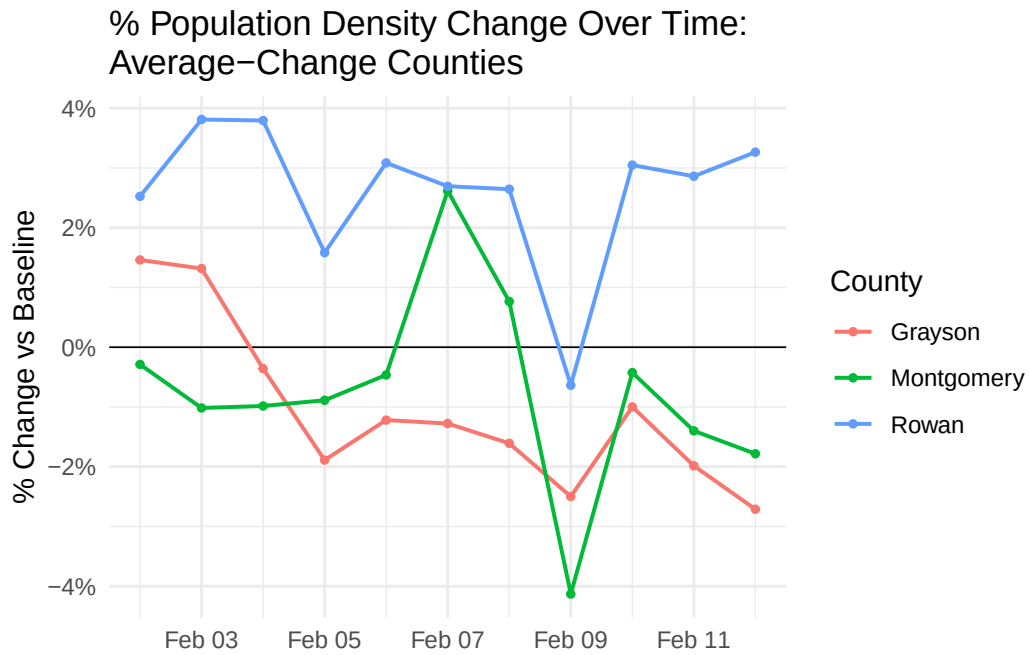


Figure 4: There are strong difference in population changes across different counties

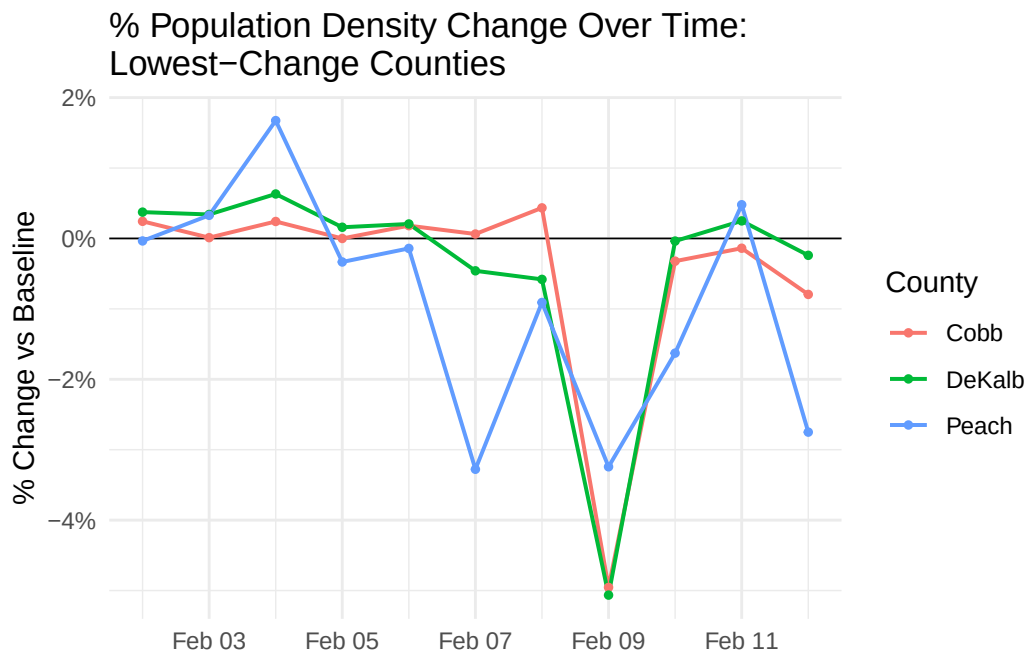


Figure 5: There are strong difference in population changes across different counties

% Population Density Change Over Time: Highest-Change Counties

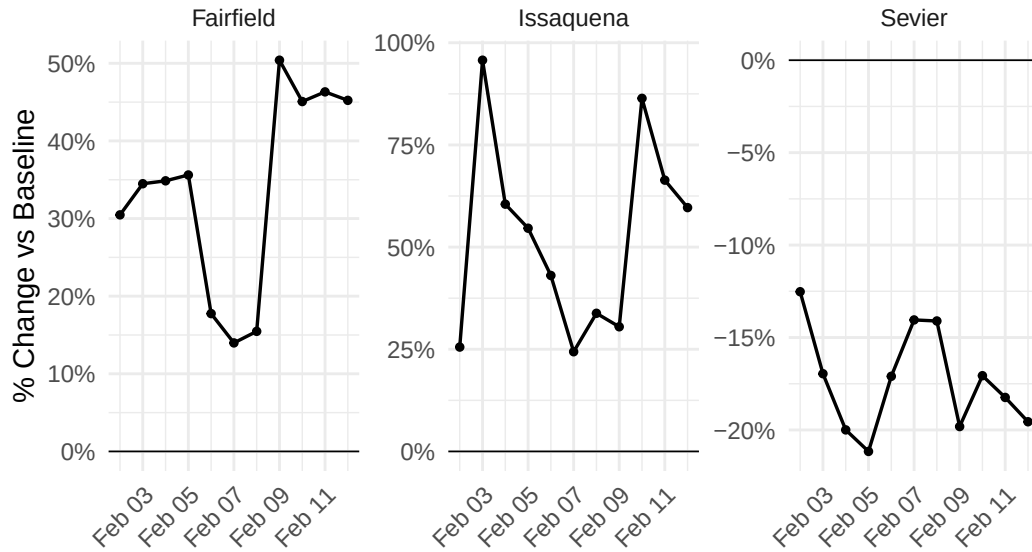


Figure 6: There are strong difference in population changes across different counties

% Population Density Change Over Time: Average-Change Counties

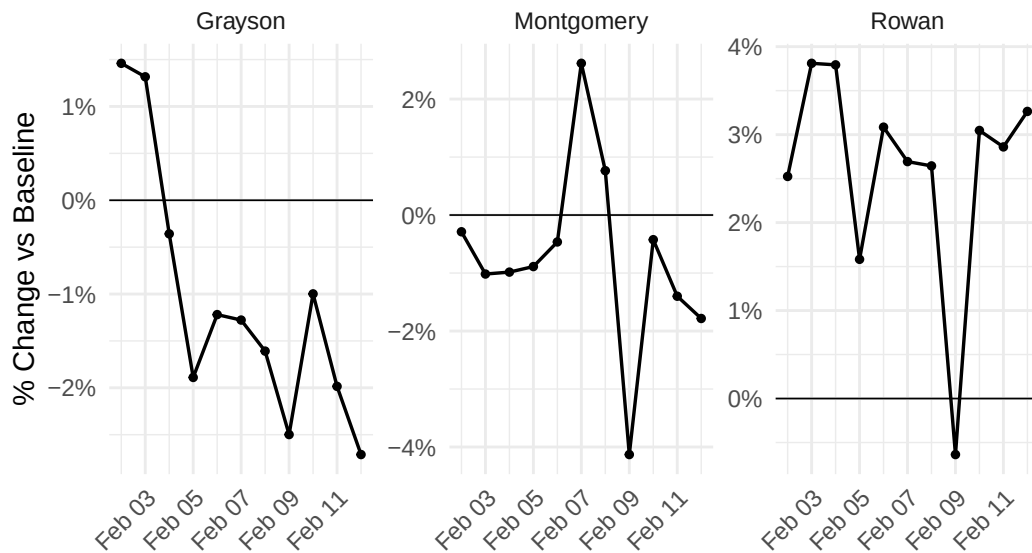


Figure 7: There are strong difference in population changes across different counties

% Population Density Change Over Time:
Lowest-Change Counties

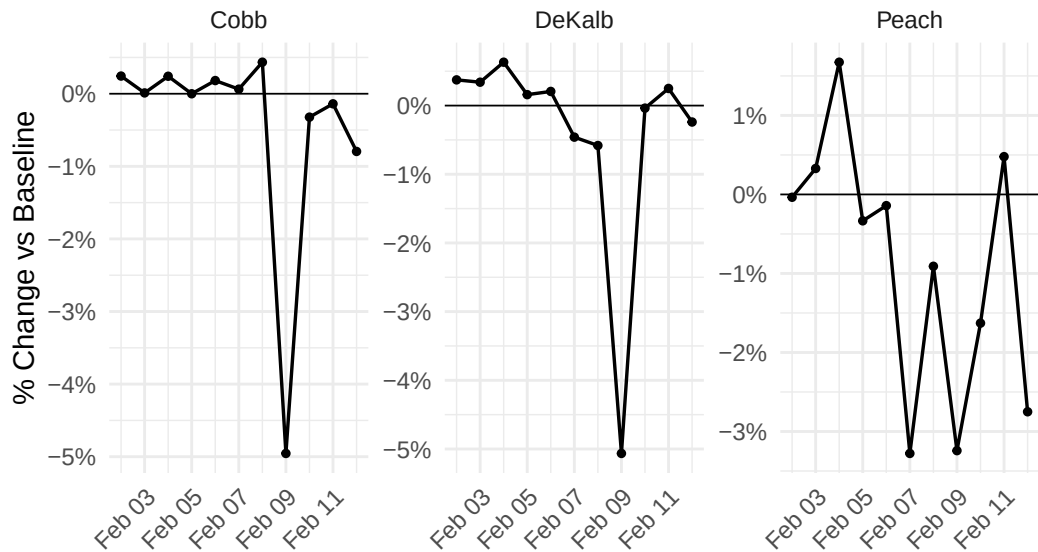


Figure 8: There are strong difference in population changes across different counties

Places

Table 1

start_county_name	end_county_name	total_crisis_movement	avg_z_score
Jefferson	Jefferson	6548192	0.1003103
Franklin	Franklin	5262301	-0.2757655
Hamilton	Hamilton	4797827	-0.2801593
Madison	Madison	4333353	-0.1587142
Shelby	Shelby	4228365	-0.2163702
Marion	Marion	4156341	-0.2564067
St. Louis	St. Louis	3805363	-0.3254192
Montgomery	Montgomery	3156012	-0.2643781
Fulton	Fulton	2608909	-0.4469157
Davidson	Davidson	2534903	0.5791654
Gwinnett	Gwinnett	2526951	-0.9734793
Knox	Knox	2223121	0.3169843
Fayette	Fayette	2059553	-0.1049316
Cobb	Cobb	1921182	-0.2391213
Greenville	Greenville	1897619	0.1128207
DeKalb	DeKalb	1868152	-0.5146540
Warren	Warren	1585751	-0.4665952
Butler	Butler	1498309	-0.1566856
Rutherford	Rutherford	1473273	-0.4580661
Richland	Richland	1442676	0.5490029

Counties with Most Increased Population Density on 2026-02-02

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Fairfield	South Carolina	30.5	20,550	\$47,885	15.6%	3.2%	5.9%
Issaquena	Mississippi	25.5	928	\$31,429	16.2%	2.7%	14.0%
Grenada	Mississippi	17.9	21,217	\$48,804	24.6%	2.3%	7.9%
Hampton	South Carolina	16.6	18,254	\$44,711	20.7%	2.2%	11.0%
York	South Carolina	14.5	293,673	\$89,095	8.4%	2.2%	3.7%
Orangeburg	South Carolina	13.8	83,253	\$45,675	22.4%	3.1%	9.0%
Chester	South Carolina	13.8	32,182	\$52,458	18.7%	2.5%	6.3%
Holmes	Mississippi	13.5	16,191	\$32,538	36.4%	1.9%	15.9%
Lincoln	North Carolina	12.7	92,716	\$80,016	11.2%	2.9%	3.3%
Lake	Tennessee	12.5	6,579	\$28,814	32.2%	2.2%	18.3%
Alexander	North Carolina	12.2	36,412	\$65,354	12.6%	2.4%	3.1%
Quitman	Mississippi	11.0	5,774	\$32,412	29.6%	2.4%	10.9%
Alexander	Illinois	10.7	4,875	\$47,043	19.0%	4.4%	10.1%
Gaston	North Carolina	10.7	234,881	\$67,478	13.5%	2.3%	5.1%
Osage	Missouri	10.6	13,402	\$76,681	8.6%	2.6%	5.6%

Counties with Most Decreased Population Density on 2026-02-02

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Sevier	Tennessee	-12.5	99,652	\$62,581	13.5%	2.7%	4.1%
Carroll	Ohio	-9.6	26,659	\$64,835	12.9%	3.5%	6.8%
Jefferson	Arkansas	-8.9	64,802	\$51,096	18.6%	2.8%	7.5%
Clay	West Virginia	-6.7	7,835	\$42,318	26.4%	3.5%	10.3%
Fannin	Georgia	-6.7	25,742	\$57,073	13.7%	2.9%	3.0%
Taliaferro	Georgia	-6.1	1,558	\$52,500	20.4%	2.9%	5.8%
Brown	Indiana	-5.4	15,606	\$78,528	7.8%	4.5%	3.5%
Swain	North Carolina	-5.4	14,013	\$52,349	19.1%	3.8%	6.6%
Taylor	Georgia	-5.2	7,786	\$41,788	32.1%	4.2%	15.5%
Greene	Alabama	-4.8	7,424	\$29,200	40.1%	4.5%	13.3%
Watauga	North Carolina	-4.5	55,123	\$51,693	25.5%	2.3%	4.4%
White	Georgia	-3.6	28,810	\$70,666	12.4%	3.2%	2.6%
Gilmer	Georgia	-3.4	32,426	\$74,499	16.0%	3.2%	2.9%
Tishomingo	Mississippi	-3.3	18,660	\$53,040	18.0%	2.7%	4.8%
Morgan	Indiana	-3.3	72,659	\$79,429	9.3%	3.2%	3.6%

Demographic Profile

Counties with Most Increased Population Density on 2026-02-12

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Issaquena	Mississippi	59.7	928	\$31,429	16.2%	2.7%	14.0%
Fairfield	South Carolina	45.2	20,550	\$47,885	15.6%	3.2%	5.9%
Osage	Missouri	12.6	13,402	\$76,681	8.6%	2.6%	5.6%
Perry	Alabama	12.6	8,031	\$37,654	31.4%	3.3%	8.0%
New Madrid	Missouri	10.6	15,731	\$51,881	17.7%	3.3%	7.2%
Norton	Virginia	10.1	3,577	\$41,495	26.3%	2.3%	10.7%
Brown	Illinois	9.9	6,322	\$67,917	9.2%	3.1%	8.5%
Humphreys	Mississippi	9.5	7,395	\$33,731	27.0%	3.1%	13.0%
Coahoma	Mississippi	9.3	20,540	\$37,461	35.5%	2.3%	14.9%
East Carroll	Louisiana	8.9	7,072	\$33,341	35.2%	1.3%	9.2%
Sumter	Alabama	8.4	11,878	\$33,310	26.5%	2.7%	9.5%
Leflore	Mississippi	7.9	27,141	\$35,277	31.5%	2.1%	14.7%
Holmes	Mississippi	7.8	16,191	\$32,538	36.4%	1.9%	15.9%
Lee	Arkansas	7.6	8,359	\$34,375	39.4%	2.3%	16.7%
Reynolds	Missouri	7.4	6,010	\$45,028	17.3%	3.8%	7.4%

Counties with Most Decreased Population Density on 2026-02-12

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Sevier	Tennessee	-19.6	99,652	\$62,581	13.5%	2.7%	4.1%
Laurens	Georgia	-15.6	49,798	\$55,010	22.6%	2.3%	6.3%
Fannin	Georgia	-13.8	25,742	\$57,073	13.7%	2.9%	3.0%
Alleghany	North Carolina	-13.8	11,174	\$47,172	16.0%	2.6%	6.0%
Clay	West Virginia	-11.3	7,835	\$42,318	26.4%	3.5%	10.3%
Gilmer	Georgia	-10.1	32,426	\$74,499	16.0%	3.2%	2.9%
Ashe	North Carolina	-9.1	26,950	\$56,866	13.5%	3.5%	4.4%
Wilkes	North Carolina	-8.7	65,935	\$51,721	16.7%	2.6%	5.8%
White	Georgia	-8.6	28,810	\$70,666	12.4%	3.2%	2.6%
Watauga	North Carolina	-8.5	55,123	\$51,693	25.5%	2.3%	4.4%
Macon	North Carolina	-8.1	37,941	\$59,061	11.6%	3.2%	6.1%
Brown	Indiana	-7.3	15,606	\$78,528	7.8%	4.5%	3.5%
Taylor	Georgia	-7.1	7,786	\$41,788	32.1%	4.2%	15.5%
Hocking	Ohio	-7.0	27,799	\$62,960	16.0%	3.3%	5.7%
Morgan	Indiana	-6.8	72,659	\$79,429	9.3%	3.2%	3.6%

Counties with Most Increased Population Density on 2026-01-30

Source: Administrative Regions

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Quitman	Mississippi	11.6	5,774	\$32,412	29.6%	2.4%	10.9%
Clarke	Georgia	10.6	129,609	\$53,625	26.0%	1.5%	6.3%
Tate	Mississippi	9.3	28,321	\$69,704	16.9%	2.7%	3.0%
Chester	Tennessee	9.1	17,611	\$59,341	16.1%	1.2%	3.1%
Tipton	Tennessee	8.1	61,553	\$74,127	13.6%	2.4%	5.2%
Oktober	Mississippi	7.4	51,771	\$46,695	25.5%	1.7%	5.4%
Fayette	Tennessee	6.5	43,267	\$88,456	10.3%	3.5%	5.3%
Union	Kentucky	6.4	13,260	\$60,327	18.4%	3.0%	5.6%
Lee	Mississippi	6.2	83,034	\$67,863	14.6%	2.1%	6.0%
Chicot	Arkansas	5.9	9,765	\$39,045	22.6%	3.6%	10.1%
Fulton	Kentucky	5.8	6,409	\$38,217	22.6%	2.5%	17.0%
Morehouse	Louisiana	5.6	24,551	\$39,454	29.3%	3.6%	11.6%
Athens	Ohio	5.4	61,737	\$56,001	21.5%	1.7%	6.8%
Weakley	Tennessee	5.0	33,016	\$51,880	19.0%	2.6%	5.5%
Monroe	Arkansas	4.9	6,589	\$43,214	20.6%	3.1%	14.5%

Counties with Most Decreased Population Density on 2026-01-30

Source: Administrative Regions

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Sevier	Tennessee	-28.3	99,652	\$62,581	13.5%	2.7%	4.1%
Sharkey	Mississippi	-20.3	3,872	\$40,000	30.9%	2.1%	8.0%
Lafayette	Mississippi	-17.5	58,327	\$67,185	18.5%	2.0%	6.7%
Tippah	Mississippi	-16.6	21,523	\$52,542	18.9%	3.6%	2.2%
Holmes	Mississippi	-15.9	16,191	\$32,538	36.4%	1.9%	15.9%
Carroll	Mississippi	-14.7	9,686	\$59,327	17.9%	1.7%	4.5%
Pope	Illinois	-13.6	3,739	\$60,050	13.3%	4.1%	5.5%
Hancock	Georgia	-13.4	8,650	\$40,082	30.6%	3.6%	8.4%
Taliaferro	Georgia	-13.2	1,558	\$52,500	20.4%	2.9%	5.8%
Swain	North Carolina	-12.7	14,013	\$52,349	19.1%	3.8%	6.6%
Fannin	Georgia	-12.4	25,742	\$57,073	13.7%	2.9%	3.0%
Alleghany	North Carolina	-12.1	11,174	\$47,172	16.0%	2.6%	6.0%
Tishomingo	Mississippi	-11.7	18,660	\$53,040	18.0%	2.7%	4.8%
Humphreys	Mississippi	-10.8	7,395	\$33,731	27.0%	3.1%	13.0%
White	Georgia	-10.0	28,810	\$70,666	12.4%	3.2%	2.6%

Counties with Most Increased Population Density on 2026-02-12

Source: Administrative Regions

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Lafayette	Mississippi	21.4	58,327	\$67,185	18.5%	2.0%	6.7%
Richland	Louisiana	16.7	19,822	\$53,544	24.0%	2.9%	8.7%
Oktibbeha	Mississippi	12.2	51,771	\$46,695	25.5%	1.7%	5.4%
Norton	Virginia	9.7	3,577	\$41,495	26.3%	2.3%	10.7%
Clarke	Georgia	9.5	129,609	\$53,625	26.0%	1.5%	6.3%
Rowan	Kentucky	8.2	24,578	\$54,321	24.2%	1.3%	8.4%
Davidson	Tennessee	6.9	715,388	\$77,853	13.9%	1.9%	6.5%
Athens	Ohio	6.5	61,737	\$56,001	21.5%	1.7%	6.8%
Grenada	Mississippi	5.5	21,217	\$48,804	24.6%	2.3%	7.9%
Panola	Mississippi	5.2	32,808	\$45,945	23.3%	2.1%	6.4%
Osage	Missouri	5.1	13,402	\$76,681	8.6%	2.6%	5.6%
Richland	South Carolina	4.9	422,117	\$63,784	17.2%	2.2%	7.2%
Christian	Kentucky	4.9	72,069	\$55,494	17.1%	1.8%	7.8%
Fulton	Georgia	4.9	1,076,561	\$95,292	12.3%	1.8%	10.0%
Fayette	Kentucky	4.8	323,725	\$69,479	14.9%	2.2%	7.2%

Counties with Most Decreased Population Density on 2026-02-12

Source: Administrative Regions

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Sevier	Tennessee	-22.8	99,652	\$62,581	13.5%	2.7%	4.1%
Pope	Illinois	-12.5	3,739	\$60,050	13.3%	4.1%	5.5%
Alleghany	North Carolina	-11.3	11,174	\$47,172	16.0%	2.6%	6.0%
Fannin	Georgia	-11.3	25,742	\$57,073	13.7%	2.9%	3.0%
Calhoun	Illinois	-11.1	4,330	\$93,203	6.3%	2.7%	2.7%
Taliaferro	Georgia	-10.6	1,558	\$52,500	20.4%	2.9%	5.8%
Brown	Indiana	-9.9	15,606	\$78,528	7.8%	4.5%	3.5%
Carroll	Mississippi	-9.0	9,686	\$59,327	17.9%	1.7%	4.5%
Gilmer	Georgia	-8.7	32,426	\$74,499	16.0%	3.2%	2.9%
Ashe	North Carolina	-8.7	26,950	\$56,866	13.5%	3.5%	4.4%
Hancock	Georgia	-8.4	8,650	\$40,082	30.6%	3.6%	8.4%
Talbot	Georgia	-8.2	5,742	\$41,597	25.2%	4.0%	9.6%
White	Georgia	-8.1	28,810	\$70,666	12.4%	3.2%	2.6%
Noble	Ohio	-7.9	14,282	\$56,098	10.6%	3.3%	7.7%
Harris	Georgia	-7.4	36,086	\$97,302	9.0%	2.8%	1.4%